Networking Module 7&8

S Vimal

1. Try Test-Connection and nslookup commands for below websites

www.google.com

www.facebook.com

www.amazon.com

www.github.com

www.cisco.com

First lets try Test-Connection command

```
PS C:\Users\vimal> Test-Connection -ComputerName www.google.com -Count 4

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-JKT... www.google.com  142.251.220.100
LAPTOP-JKT... www.google.com  142.251.220.100
LAPTOP-JKT... www.google.com  142.251.220.100
LAPTOP-JKT... www.google.com  142.251.220.100


PS C:\Users\vimal> Test-Connection -ComputerName www.facebook.com -Count 4

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-JKT... www.facebook... 57.144.42.1
LAPTOP-JKT... www.facebook... 57.144.42.1
LAPTOP-JKT... www.facebook... 57.144.42.1
LAPTOP-JKT... www.facebook... 57.144.42.1


PS C:\Users\vimal> Test-Connection -ComputerName www.amazon.com -Count 4

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-JKT... www.amazon.com  18.161.240.42
LAPTOP-JKT... www.amazon.com  18.161.240.42
LAPTOP-JKT... www.amazon.com  18.161.240.42
LAPTOP-JKT... www.amazon.com  18.161.240.42
```

```
PS C:\Users\vimal> Test-Connection -ComputerName www.github.com -Count 4

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-JKT... www.github.com  20.207.73.82
LAPTOP-JKT... www.github.com  20.207.73.82
LAPTOP-JKT... www.github.com  20.207.73.82
LAPTOP-JKT... www.github.com  20.207.73.82


PS C:\Users\vimal> Test-Connection -ComputerName www.cisco.com -Count 4

Source        Destination     IPV4Address      IPV6Address
------        -----------     -----------      -----------
LAPTOP-JKT... www.cisco.com   23.208.176.119
LAPTOP-JKT... www.cisco.com   23.208.176.119
LAPTOP-JKT... www.cisco.com   23.208.176.119
LAPTOP-JKT... www.cisco.com   23.208.176.119
```

nslookup command

```
PS C:\Users\vimal> nslookup www.google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482c:7700::
          2001:4860:4828:7700::
          2001:4860:4829:7700::
          2001:4860:4826:7700::
          2001:4860:482b:7700::
          2001:4860:4827:7700::
          2001:4860:482d:7700::
          2001:4860:482a:7700::
          142.251.151.119
          142.251.152.119
          142.251.154.119
          142.251.150.119
          142.251.156.119
          142.251.157.119
          142.251.155.119
          142.251.153.119

PS C:\Users\vimal> nslookup www.facebook.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    star-mini.c10r.facebook.com
Addresses:  2a03:2880:f334:1:face:b00c:0:25de
          57.144.42.1
Aliases:  www.facebook.com
```

```
PS C:\Users\vimal> nslookup www.amazon.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    cf.47cf2c8c9-frontier.amazon.com
Addresses:  2600:9000:257a:fa00:7:49a5:5fd6:da1
          2600:9000:257a:8e00:7:49a5:5fd6:da1
          2600:9000:257a:a200:7:49a5:5fd6:da1
          2600:9000:257a:e600:7:49a5:5fd6:da1
          2600:9000:257a:da00:7:49a5:5fd6:da1
          2600:9000:257a:1000:7:49a5:5fd6:da1
          2600:9000:257a:2400:7:49a5:5fd6:da1
          2600:9000:257a:e800:7:49a5:5fd6:da1
          18.161.240.42
Aliases:  www.amazon.com
          tp.47cf2c8c9-frontier.amazon.com

PS C:\Users\vimal> nslookup www.github.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    github.com
Address:  20.207.73.82
Aliases:  www.github.com

PS C:\Users\vimal> nslookup www.cisco.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:    e2867.dsca.akamaiedge.net
Addresses:  2600:140f:5e00:480::b33
          2600:140f:5e00:482::b33
          23.208.176.119
Aliases:  www.cisco.com
          www.cisco.com.akadns.net
          wwwds.cisco.com.edgekey.net
          wwwds.cisco.com.edgekey.net.globalredir.akadns.net
```

Test-Connection is the PowerShell equivalent of ping. It sends ICMP echo request packets to a target host and waits for replies, confirming whether the host is reachable over the network. Key parameters: -ComputerName specifies the target hostname or IP, -Count sets how many packets to send. The output shows the source machine, the destination, the resolved IPv4 address, and optionally response time and TTL.

NSLookup (Name Server Lookup) queries the configured DNS server to resolve a hostname into its IP address. It reveals not just the final IP but also CNAME alias chains, whether the answer is authoritative or cached, and both IPv4/IPv6 addresses.

2. Use Wireshark to capture and analyze DNS, TCP, UDP traffic and packet header,

packet flow, options and flags

For this experimentation, I started wireshark capture and entered www.google.com and then stopped the capture.

DNS captures, request and its response packet

173	8.980826200	172.20.24.176	8.8.8.8	DNS	85
179	8.984867100	8.8.8.8	172.20.24.176	DNS	136
194	8.989958800	172.20.24.176	8.8.8.8	DNS	75
201	8.994403800	8.8.8.8	172.20.24.176	DNS	91
224	9.006888600	8.8.8.8	172.20.24.176	DNS	93

```
▸ Frame 173: Packet, 85 bytes on wire (680 bits), 85 bytes captured (680 bit
▸ Ethernet II, Src: CloudNetwork_ea:67:b5 (14:ac:60:ea:67:b5), Dst: Cisco_61
▸ Internet Protocol Version 4, Src: 172.20.24.176, Dst: 8.8.8.8
▸ User Datagram Protocol, Src Port: 53340, Dst Port: 53
▾ Domain Name System (query)
    Transaction ID: 0x3679
  ▸ Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
    [Response In: 179]
```

179	8.984867100	8.8.8.8	172.20.24.176	DNS	130
194	8.989958800	172.20.24.176	8.8.8.8	DNS	75
201	8.994403800	8.8.8.8	172.20.24.176	DNS	91
224	9.006888600	8.8.8.8	172.20.24.176	DNS	93

```
▸ Frame 179: Packet, 130 bytes on wire (1040 bits), 130 bytes captured (1040
▸ Ethernet II, Src: Cisco_61:f5:1f (68:e5:9e:61:f5:1f), Dst: CloudNetwork_ea
▸ Internet Protocol Version 4, Src: 8.8.8.8, Dst: 172.20.24.176
▸ User Datagram Protocol, Src Port: 53, Dst Port: 53340
▾ Domain Name System (response)
    Transaction ID: 0x3679
  ▸ Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
  ▸ Queries
  ▸ Answers
    [Request In: 173]
    [Time: 4.040900 milliseconds]
```

On analyzing:

Frame Layer : Has details like length of the packet, packet arrival time, time shift from previously arrived packet, packet number, protocols in the frame, encoding etc..

Ethernet: Contains source and destination MAC address.

IP: Contains the source and destination IP address and also specifies the IPv4 version, and has flags like 0x0 and sets fragment offset, time to live, the protocol used.

UDP layer: The laptop sends from an ephemeral source port to destination port 53, which is the standard DNS port. UDP length covers the header plus the DNS payload. No connection setup is needed,UDP fires the query directly without handshaking, making it faster.

DNS layer: The flags indicate it is a standard query, with 1 question and 0 replies (the request packet) and the response packet contains 1 question and 2 replies

<u>TCP capture</u>

168	8.977675600	172.20.24.176	142.251.152.119	TCP	54 58625 → 443 [ACK
169	8.978850300	142.251.152.119	172.20.24.176	TLSv1.3	1514 Application Data
170	8.980026000	142.251.152.119	172.20.24.176	TLSv1.3	1514 Application Data
171	8.980089400	172.20.24.176	142.251.152.119	TCP	54 58625 → 443 [ACK
174	8.981150700	142.251.152.119	172.20.24.176	TLSv1.3	1514 Application Data
175	8.982452500	142.251.152.119	172.20.24.176	TLSv1.3	1514 Application Data
176	8.982505200	172.20.24.176	142.251.152.119	TCP	54 58625 → 443 [ACK
177	8.983743100	142.251.152.119	172.20.24.176	TLSv1.3	1514 Application Data
178	8.984867100	142.251.152.119	172.20.24.176	TLSv1.3	1514 Application Data
180	8.984936000	172.20.24.176	142.251.152.119	TCP	54 58625 → 443 [ACK
181	8.986188800	142.251.152.119	172.20.24.176	TLSv1.3	1514 Application Data

```
▸ Frame 168: Packet, 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \De
▸ Ethernet II, Src: CloudNetwork_ea:67:b5 (14:ac:60:ea:67:b5), Dst: Cisco_61:f5:1f (68:e5:9e:61
▸ Internet Protocol Version 4, Src: 172.20.24.176, Dst: 142.251.152.119
▾ Transmission Control Protocol, Src Port: 58625, Dst Port: 443, Seq: 14760, Ack: 62768, Len: 0
    Source Port: 58625
    Destination Port: 443
    [Stream index: 6]
    [Stream Packet Number: 105]
  ▸ [Conversation completeness: Incomplete, DATA (15)]
    [TCP Segment Len: 0]
    Sequence Number: 14760    (relative sequence number)
    Sequence Number (raw): 3600176608
    [Next Sequence Number: 14760    (relative sequence number)]
    Acknowledgment Number: 62768    (relative ack number)
    Acknowledgment number (raw): 2901819727
    0101 .... = Header Length: 20 bytes (5)
  ▸ Flags: 0x010 (ACK)
    Window: 255
    [Calculated window size: 65280]
```

Frame: Has details like length of the packet, packet arrival time, time shift from previously arrived packet, packet number, protocols in the frame, encoding etc..

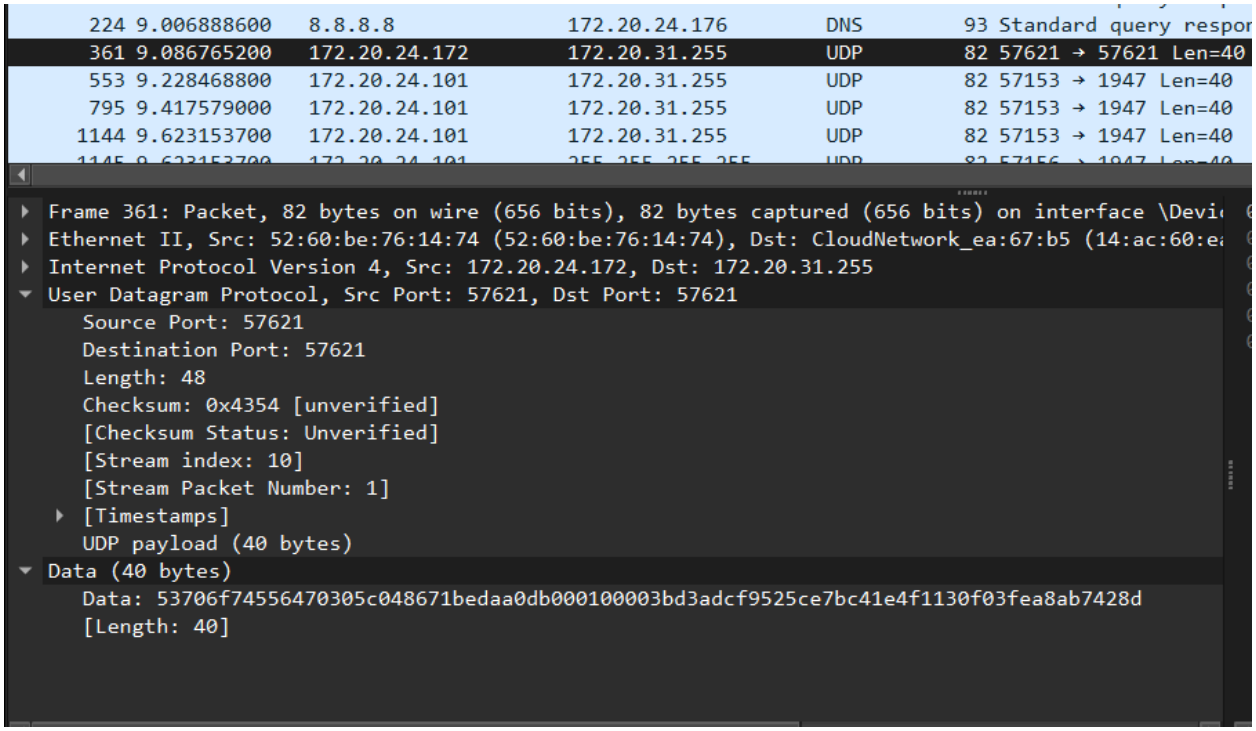
Ethernet: Contains source and destination MAC address.

IP: Contains the source and destination IP address and also specifies the IPv4 version, and has flags like 0x2 which means do not fragment

TCP: The laptop (port 58625) acknowledging data from Google (port 443 = HTTPS). Seq=14760 tracks bytes sent, Ack=62768 confirms bytes received from Google. Only the

ACK flag is set with zero data length, this is a pure acknowledgment packet, no payload. Window=65280 tells the destination how much data the laptop can still accept.

<u>UDP Capture</u>



224	9.006888600	8.8.8.8	172.20.24.176	DNS	93	Standard query respor
361	9.086765200	172.20.24.172	172.20.31.255	UDP	82	57621 → 57621 Len=40
553	9.228468800	172.20.24.101	172.20.31.255	UDP	82	57153 → 1947 Len=40
795	9.417579000	172.20.24.101	172.20.31.255	UDP	82	57153 → 1947 Len=40
1144	9.623153700	172.20.24.101	172.20.31.255	UDP	82	57153 → 1947 Len=40

```
Frame 361: Packet, 82 bytes on wire (656 bits), 82 bytes captured (656 bits) on interface \Devi
Ethernet II, Src: 52:60:be:76:14:74 (52:60:be:76:14:74), Dst: CloudNetwork_ea:67:b5 (14:ac:60:e
Internet Protocol Version 4, Src: 172.20.24.172, Dst: 172.20.31.255
User Datagram Protocol, Src Port: 57621, Dst Port: 57621
    Source Port: 57621
    Destination Port: 57621
    Length: 48
    Checksum: 0x4354 [unverified]
    [Checksum Status: Unverified]
    [Stream index: 10]
    [Stream Packet Number: 1]
    [Timestamps]
    UDP payload (40 bytes)
Data (40 bytes)
    Data: 53706f74556470305c048671bedaa0db000100003bd3adcf9525ce7bc41e4f1130f03fea8ab7428d
    [Length: 40]
```

UDP layer:

Frame: Has details like length of the packet, packet arrival time, time shift from previously arrived packet, packet number, protocols in the frame, encoding as ASCII

Ethernet: Contains source and destination MAC address.

IP: Contains the source and destination IP address and also specifies the IPv4 version, and has flags like 0x2 which means do not fragment

UDP: Has details like the source and destination port, payload is 40 bytes of raw binary data with no DNS or other structured protocol on top

3. Explore traceroute/tracert for different websites eg:google.com and analyse the

parameters in the output and explore different options for traceroute command

```
PS C:\Users\vimal> tracert google.com

Tracing route to google.com [142.251.43.142]
over a maximum of 30 hops:

  1     2 ms     2 ms     1 ms  172.20.24.1
  2     2 ms     2 ms     2 ms  172.20.72.100
  3     3 ms     3 ms     3 ms  122.184.65.193
  4     4 ms     3 ms     3 ms  116.119.164.19
  5     5 ms     5 ms     4 ms  72.14.205.196
  6     4 ms    55 ms     4 ms  216.239.43.131
  7     3 ms     3 ms     3 ms  142.250.236.157
  8     4 ms     4 ms     4 ms  bkk03s01-in-f14.1e100.net [142.251.43.142]

Trace complete.
PS C:\Users\vimal> tracert www.facebook.com

Tracing route to star-mini.c10r.facebook.com [57.144.56.1]
over a maximum of 30 hops:

  1     2 ms     1 ms     1 ms  172.20.24.1
  2     2 ms     2 ms     2 ms  172.20.72.100
  3     3 ms     3 ms     3 ms  122.184.65.193
  4     *        4 ms     3 ms  116.119.164.19
  5     5 ms     3 ms     4 ms  ae5.pr01.maa1.tfbnw.net [157.240.68.40]
  6     4 ms     4 ms     4 ms  po4004.asw01.maa2.tfbnw.net [129.134.41.106]
  7     4 ms     4 ms     4 ms  usw01.maa5.tfbnw.net [129.134.118.19]
  8     7 ms     7 ms     5 ms  163.77.246.102
  9     4 ms     4 ms     4 ms  edge-star-mini-shv-02-maa5.facebook.com [57.144.56.1]
```

The output is of the form:

hop#count RTT1 RTT2 RTT3 IP

Hopcount : shows the count of that hop

RTT1/2/3: Roundtrip time for 3 probes

IP: IP of the router at that hop

Other tracert options include

a. **tracert -h 3 www.google.com - stop at 3 hops, in this case stops before reaching**

```
PS C:\Users\vimal> tracert -h 3 www.google.com

Tracing route to www.google.com [142.251.151.119]
over a maximum of 3 hops:

  1     2 ms     1 ms     1 ms  172.20.24.1
  2     2 ms     2 ms     2 ms  172.20.72.100
  3     3 ms     3 ms     3 ms  122.184.65.193
```

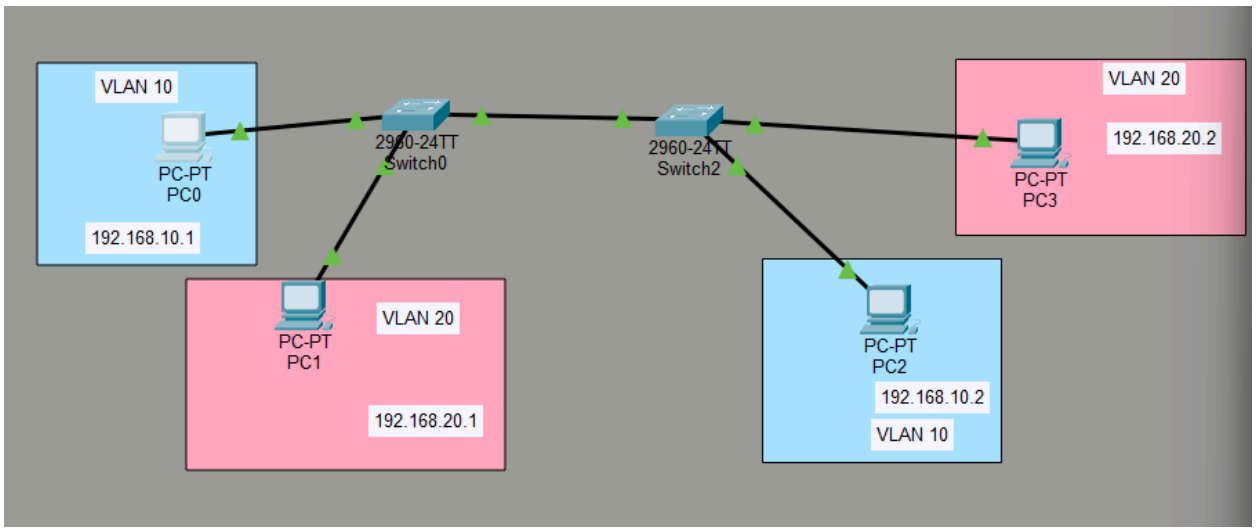
b. **tracert -w 500 www.cisco.com - shorter timeout per hop**

```
PS C:\Users\vimal> tracert -w 50 www.cisco.com

Tracing route to e2867.dsca.akamaiedge.net [23.61.113.60]
over a maximum of 30 hops:

  1     2 ms     2 ms     2 ms  172.20.24.1
  2     2 ms     3 ms     2 ms  172.20.72.100
  3     4 ms     4 ms     4 ms  122.184.65.193
  4    10 ms    11 ms    10 ms  116.119.73.240
  5    11 ms    10 ms    10 ms  a23-61-113-60.deploy.static.akamaitechnologies.com [23.61.113.60]
```

4. Use Cisco packet tracer for the below

5. Set up trunk ports between switches and try ping between different VLANs.

Cisco packet tracer setup



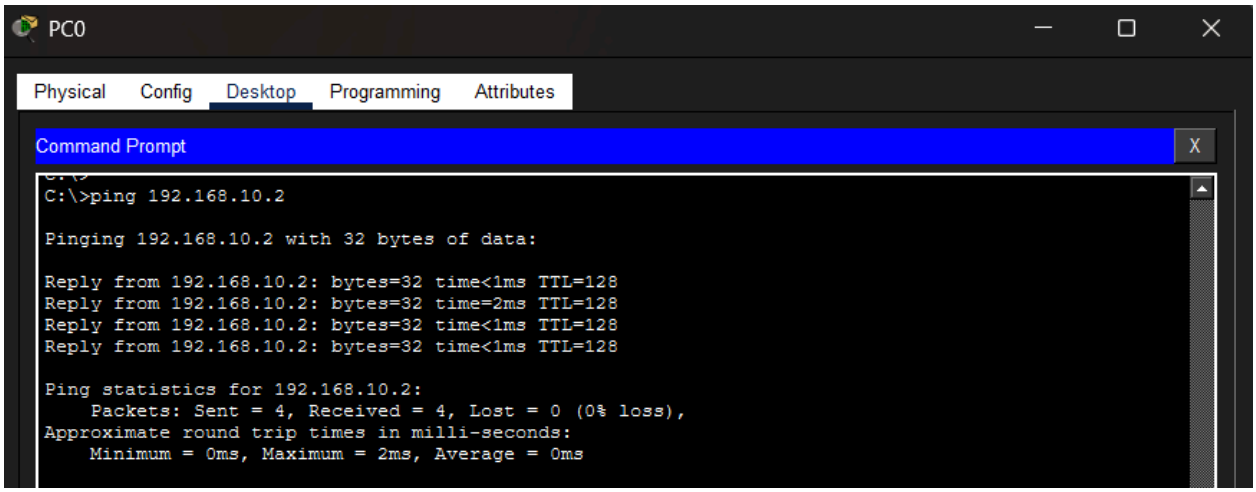
Configuring VLAN setup in switches same repeated on both switches

```
Switch(config)#vlan 10
Switch(config-vlan)#name DATA
Switch(config-vlan)#vlan 20
Switch(config-vlan)#name VOICE
Switch(config-vlan)#exit
```

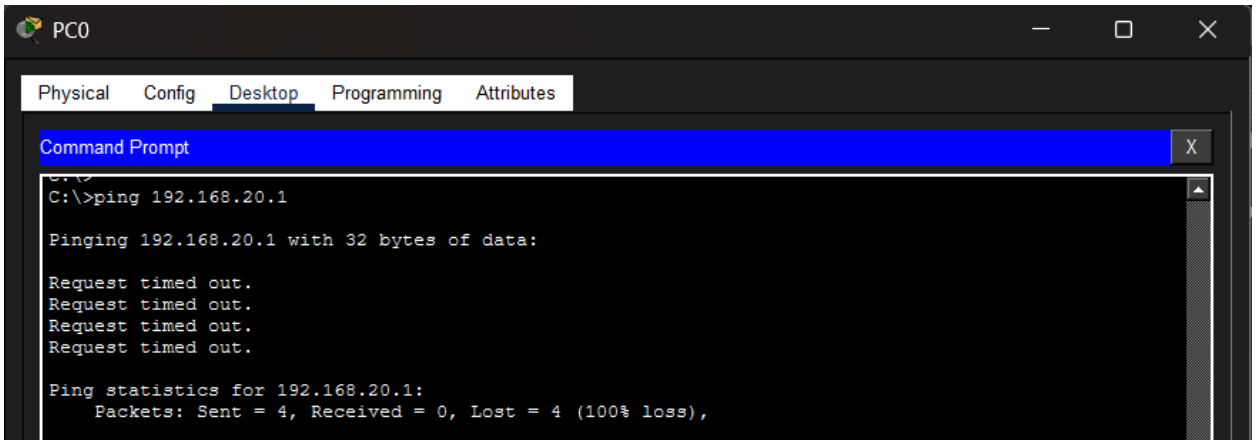
Trunk Configuration on Switch 0, same is repeated on Switch 1

```
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 10,20
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
write memory
Building configuration...
[OK]
Switch#show vlan interfaces trunk
                    ^
% Invalid input detected at '^' marker.

Switch#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/3, Fa0/4, Fa0/5, Fa0/6
                                                Fa0/7, Fa0/8, Fa0/9, Fa0/10
                                                Fa0/11, Fa0/12, Fa0/13, Fa0/14
                                                Fa0/15, Fa0/16, Fa0/17, Fa0/18
                                                Fa0/19, Fa0/20, Fa0/21, Fa0/22
                                                Fa0/23, Fa0/24, Gig0/2
10   data                             active    Fa0/1
20   voice                            active    Fa0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20

Port        Vlans allowed and active in management domain
Gig0/1      10,20
```

Ping test within same VLAN devices across different switches



Ping test between devices of different VLAN within same switch



6. Change the native VLAN on a trunk port-Test for VLAN mismatches and Troubleshoot.

Changing native vlan on switch 0

```
Switch>enable
Switch#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#switchport trunk native vlan 99
Switch(config-if)#exit
Switch(config)#
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (99), with
Switch GigabitEthernet0/1 (1).

%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (99), with
Switch GigabitEthernet0/1 (1).
```

A warning message has been seen on switch 2 also after changing the native VLAN to 99

```
%CDP-4-NATIVE_VLAN_MISMATCH: Native VLAN mismatch discovered on GigabitEthernet0/1 (1), with Switch
GigabitEthernet0/1 (99).
```

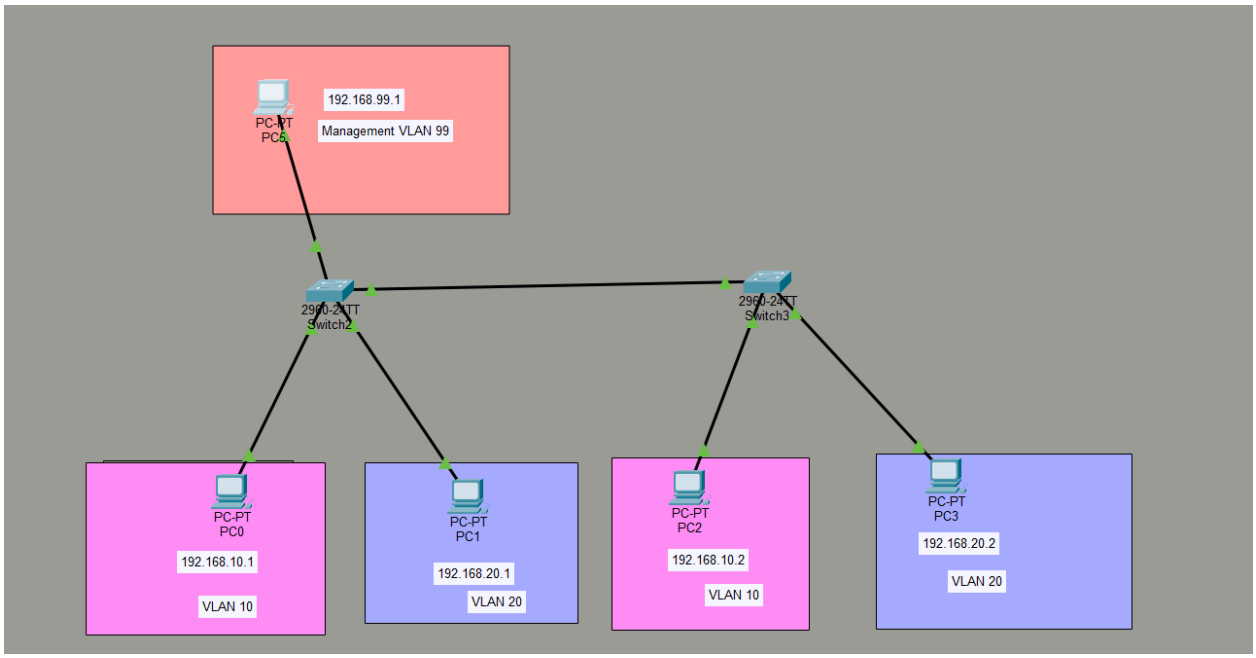
While trouble shooting we check the trunk interfaces, by default native trunk will vlan 1, but I changed this on switch 0 to 99, that change is reflected in this command

```
Switch#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      99

Port        Vlans allowed on trunk
Gig0/1      10,20

Port        Vlans allowed and active in management domain
Gig0/1      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20
```

Now I changed the default VLAN back to 1 for switch 0, this can be verified by checking the trunk interface

```
Switch(config)#interface GigabitEthernet0/1
Switch(config-if)#switchport trunk native vlan 1
Switch(config-if)#exiit
                     ^
% Invalid input detected at '^' marker.

Switch(config-if)#ex
Switch(config)#ex
Switch#
%SYS-5-CONFIG_I: Configured from console by console
show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      10,20

Port        Vlans allowed and active in management domain
Gig0/1      10,20

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      10,20
```

7. Configure a management VLAN and assign an IP address for remote access.Test SSH or Telnet access to the switch.



Here I created a new management VLAN for switch 0 with a new PC and configured default gateway and assigned VLAN 99 a port

```
Switch(config)#vlan 99
Switch(config-vlan)#name management
Switch(config-vlan)#exit
Switch(config)#interface vlan 99
```

```
Switch(config-if)#no ip address
Switch(config-if)#ip address 192.168.99.1 255.255.255.0
Switch(config-if)#no shutdown
Switch(config-if)#exit
```

```
Switch(config)#ip default-gateway 192.168.99.1
Switch(config)#
Switch(config)#interface FastEthernet0/24
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 99
```

Configuring telnet

```
Switch(config)#line vty 0 4
Switch(config-line)#password cisco123
Switch(config-line)#login
Switch(config-line)#transport input telnet
Switch(config-line)#ex
% Ambiguous command: "ex"
Switch(config-line)#exit
Switch(config)#end
```

Verifying telnet

```
C:\>telnet 192.168.99.1
Trying 192.168.99.1 ...Open


User Access Verification

Password:
Switch>
```

Configuring SSH:

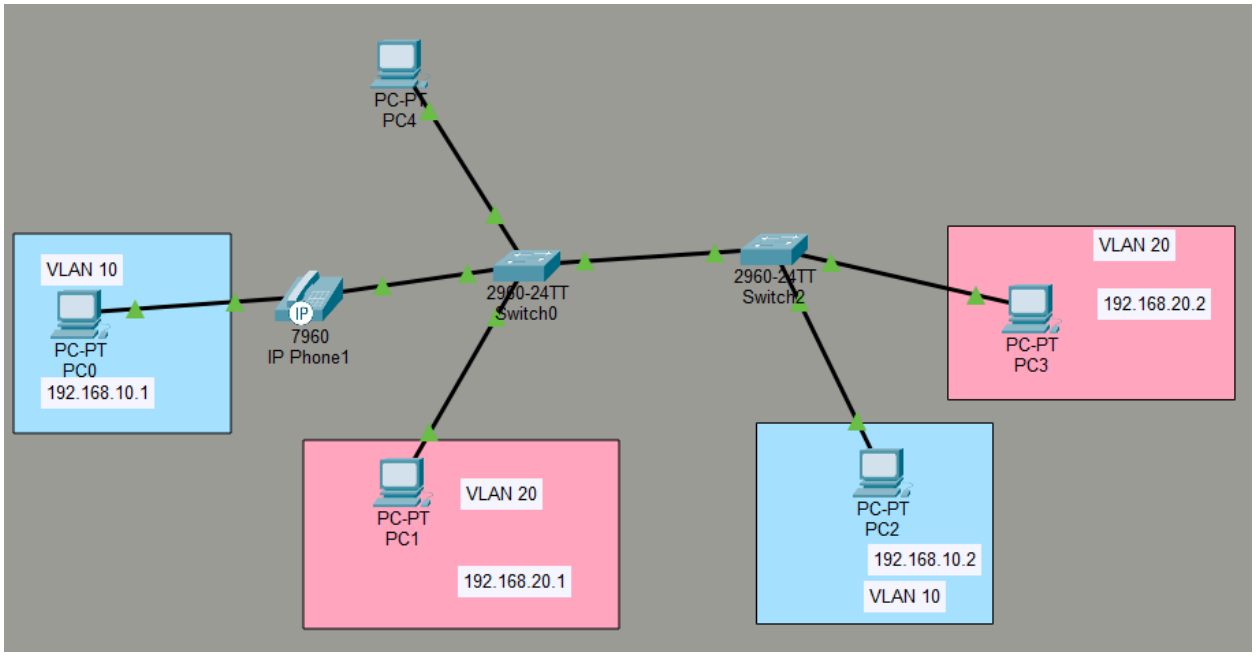
```
Switch(config)#hostname SW0
SW0(config)#ip domain-name lab.local
SW0(config)#crypto key generate rsa
The name for the keys will be: SW0.lab.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

SW0(config)#ip ssh version 2
*Mar 1 1:22:1.130: %SSH-5-ENABLED: SSH 1.99 has been enabled
SW0(config)#username admin privilege 15 secret Cisco@123
SW0(config)#line vty
% Incomplete command.
SW0(config)#line vty 0 4
SW0(config-line)#transport input ssh
SW0(config-line)#login local
```

Verifying SSH connection: The connection can be verified by running the switch functions such as show users.

```
C:\>ssh -l admin 192.168.99.1

Password:



SW0#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
SW0#show users
    Line       User       Host(s)              Idle       Location
   0 con 0                 idle                 00:01:38
*  2 vty 0     admin      idle                 00:00:00

  Interface    User               Mode         Idle     Peer Address
```

8. You have a Cisco switch and a VoIP phone that needs to be placed in a voice VLAN (VLAN 20). The data for the PC should remain in a separateVLAN (VLAN 10). Configure the switch port to support both voice and data traffic.

Configuring the switch for a VoIP between PC0 and switch0




```
SW0(config)#interface FastEthernet0/1
SW0(config-if)#switchport mode access
SW0(config-if)#switchport access vlan 10
SW0(config-if)#switchport voice vlan 20
```

Verified the configuration

```
SW0#show interfaces FastEthernet0/1 switchport
Name: Fa0/1
Switchport: Enabled
Administrative Mode: static access
Operational Mode: down
Administrative Trunking Encapsulation: dot1q
Operational Trunking Encapsulation: native
Negotiation of Trunking: Off
Access Mode VLAN: 10 (data)
Trunking Native Mode VLAN: 1 (default)
Voice VLAN: 20
Administrative private-vlan host-association: none
Administrative private-vlan mapping: none
Administrative private-vlan trunk native VLAN: none
Administrative private-vlan trunk encapsulation: dot1q
Administrative private-vlan trunk normal VLANs: none
Administrative private-vlan trunk private VLANs: none
Operational private-vlan: none
Trunking VLANs Enabled: All
Pruning VLANs Enabled: 2-1001
Capture Mode Disabled
Capture VLANs Allowed: ALL
Protected: false
```

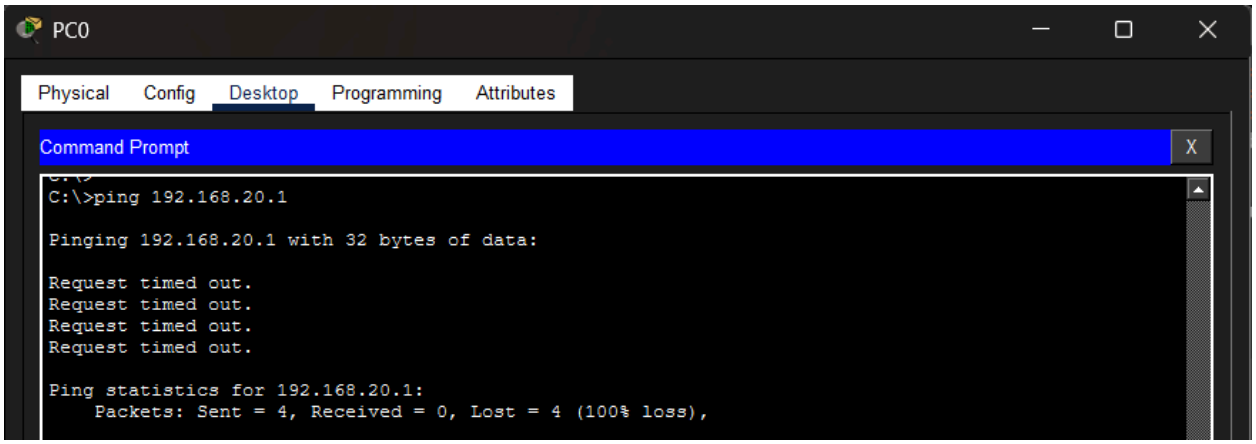
A single switch port was configured to carry both data and voice traffic simultaneously. The PC connected behind the IP phone sends untagged traffic which the switch places into VLAN 10 (data). The IP phone itself tags its own traffic with VLAN 20 (voice). before sending it to the switch. The switch separates them automatically — untagged frames go to VLAN 10 and frames tagged with VLAN 20 go to the voice VLAN. This allows one physical cable and one switch port to serve both a PC and a VoIP phone simultaneously without mixing their traffic.

9. You configured VLANs 10 and 20 on your switch and assigned ports to each VLAN.

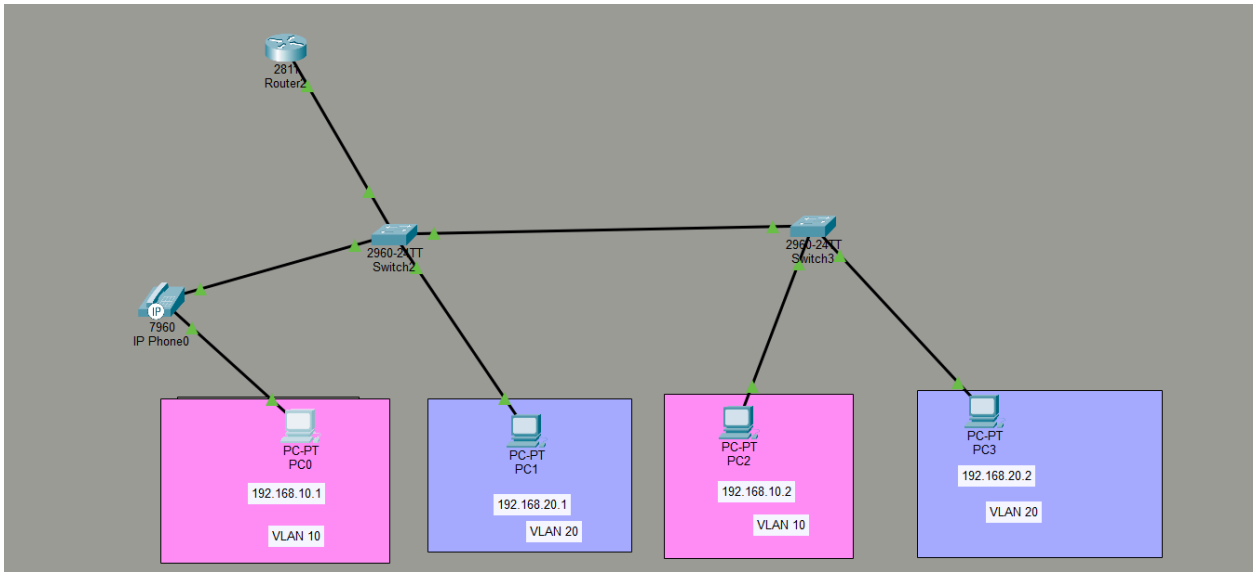
However, devices in VLAN 10 cannot communicate with devices in VLAN 20.

Troubleshoot the issue.

As we have already seen devices in VLAN 10 cannot communicate with devices in VLAN 20. This is because VLANS are separate broadcast domains. Devices in different VLANs cannot communicate at Layer 2. A Layer 3 devices such as router is required for communication between different VLANs and this process is called Inter-VLAN routing.



10. Try Inter VLAN routing with Router



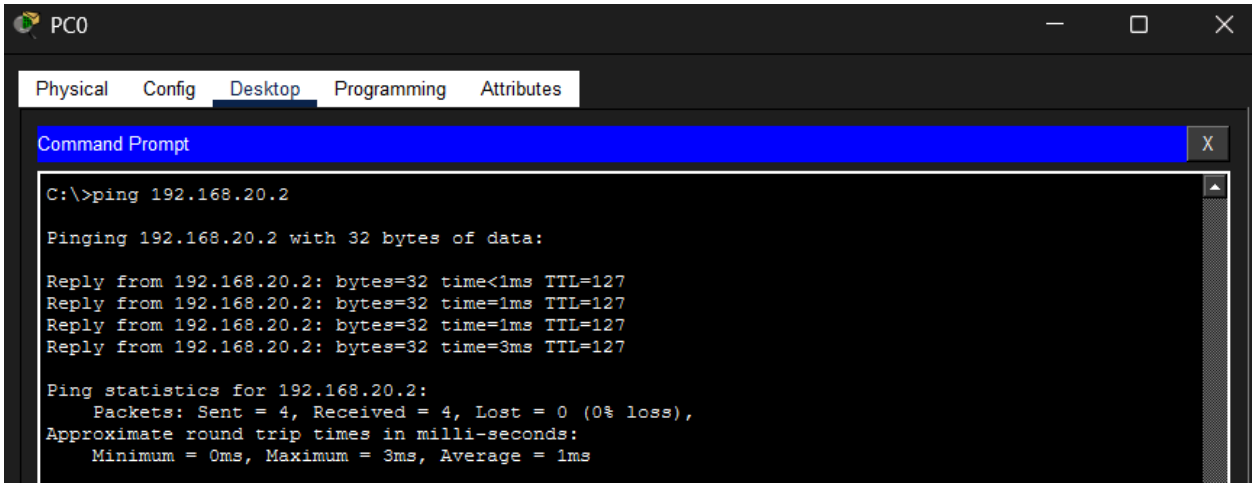
Switch configuration for Inter VLAN routing

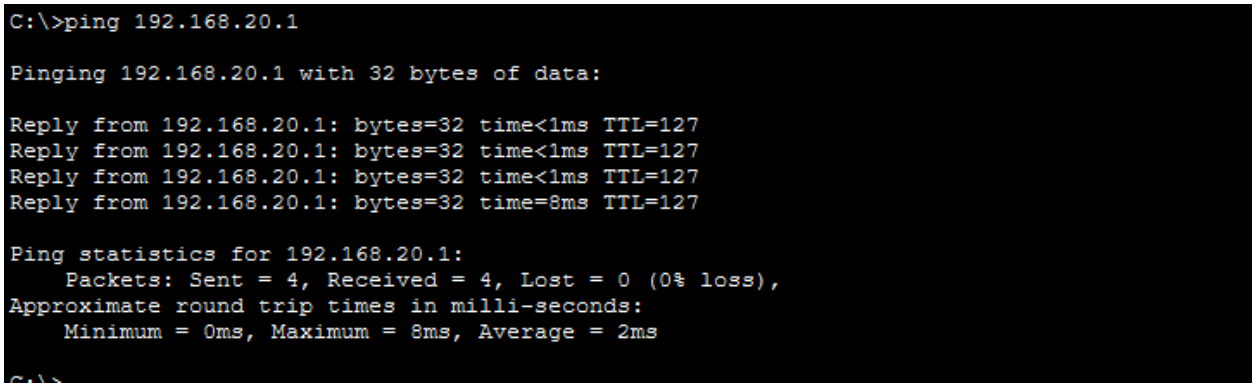
```
SW0>enable
SW0#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
SW0(config)#interface FastEthernet0/5
SW0(config-if)#switchport mode trunk
SW0(config-if)#switchport trunk allowed vlan 10,20
SW0(config-if)#
```

Router Configuration for Inter VLAN routing - Creating subinterfaces on the interface from switch and configuring the gateways of respective VLANs

```
Router(config)#interface FastEthernet0/0.10
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.10, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.10, changed state to
up

Router(config-subif)#encapsulation dot1Q 10
Router(config-subif)#ip address 192.168.20.50
% Incomplete command.
Router(config-subif)#ip address 192.168.10.50 255.255.255.0
Router(config-subif)#exit
Router(config)#interface FastEthernet0/0.20
Router(config-subif)#
%LINK-3-UPDOWN: Interface FastEthernet0/0.20, changed state to down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0.20, changed state to
up

Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.20.50 255.255.255.0
Router(config-subif)#end
```

Verifying the connections : Pinged from PC0 (192.168.10.1) of VLAN 10 to PC3 (192.168.20.2) of VLAN 20 and PC1 (192.168.20.1) VLAN 20





11. Implement ACLs to restrict traffic based on source and destination ports.Test rules by simulating legitimate and unauthorized traffic.

ACLs filter based on source IP, destination IP, protocol, port numbers. Here I will block telnet from VLAN 10 to VLAN 20 rest all will work fine.

Configuring ACL on router

```
Router>enable
Router#conf term
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#ip access-list extended BLOCK_TELNET
Router(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 eq 23
% Incomplete command.
Router(config-ext-nacl)#deny tcp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 eq 23
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
Router(config)#interface FastEthernet0/0.10
Router(config-subif)#ip access-group BLOCK_TELNET in
Router(config-subif)#exit
```

Ping from PC0(192.168.10.1) to PC3(192.168.20.2) (VLAN10 to VLAN 20) - Legitimate traffic

```
C:\>ping 192.168.20.2

Pinging 192.168.20.2 with 32 bytes of data:

Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Telnet from PC0(192.168.10.1) to PC3(192.168.20.2) - Illegal Traffic

```
C:\>telnet 192.168.20.2
Trying 192.168.20.2 ...
% Connection timed out; remote host not responding
```

12. Configure a standard Access Control List (ACL) on a router to permit traffic from a specific IP range.Test connectivity to verify the ACL is working as intended.

We have IP addresses in range 192.168.10.1 and 192.168.10.2 in VLAN10 and 192.168.20.1 and 192.168.20.2 in VLAN20, now we will restrict the computers in VLAN10 from using telnet.

First configuring the router to create an ACL

```
Router(config)#access-list 10 permit 192.168.10.0 0.0.0.255
Router(config)#access-list 10 deny any
```

Applying the ACL on telnet and configuring password

```
Router(config)#line vty 0 4
Router(config-line)#access-class 10 in
Router(config-line)#password cisco
Router(config-line)#login
Router(config-line)#exit
```

Verifying from router

```
Router#show ip access-lists
Extended IP access list BLOCK_TELNET
    10 deny tcp 192.168.10.0 0.0.0.255 192.168.20.0 0.0.0.255 eq telnet (12 match(es))
    20 permit ip any any (38 match(es))
Standard IP access list 10
    10 permit 192.168.10.0 0.0.0.255 (2 match(es))
    20 deny any (4 match(es))
```

Verifying from VLAN10 - Legitimate

```
C:\>telnet 192.168.10.50
Trying 192.168.10.50 ...Open


User Access Verification

Password:
Router>
```

```
Router>show users
    Line       User       Host(s)              Idle       Location
   0 con 0                 idle                 00:09:41
*324 vty 0                 idle                 00:00:00 192.168.10.2

  Interface    User               Mode         Idle     Peer Address
```

Verifying from VLAN20 - Illegal

```
C:\>telnet 192.168.20.50
Trying 192.168.20.50 ...
% Connection refused by remote host
C:\>
```

13. Create an extended ACL to block specific applications, such as HTTP or FTP traffic.Test the ACL rules by attempting to access blocked services.

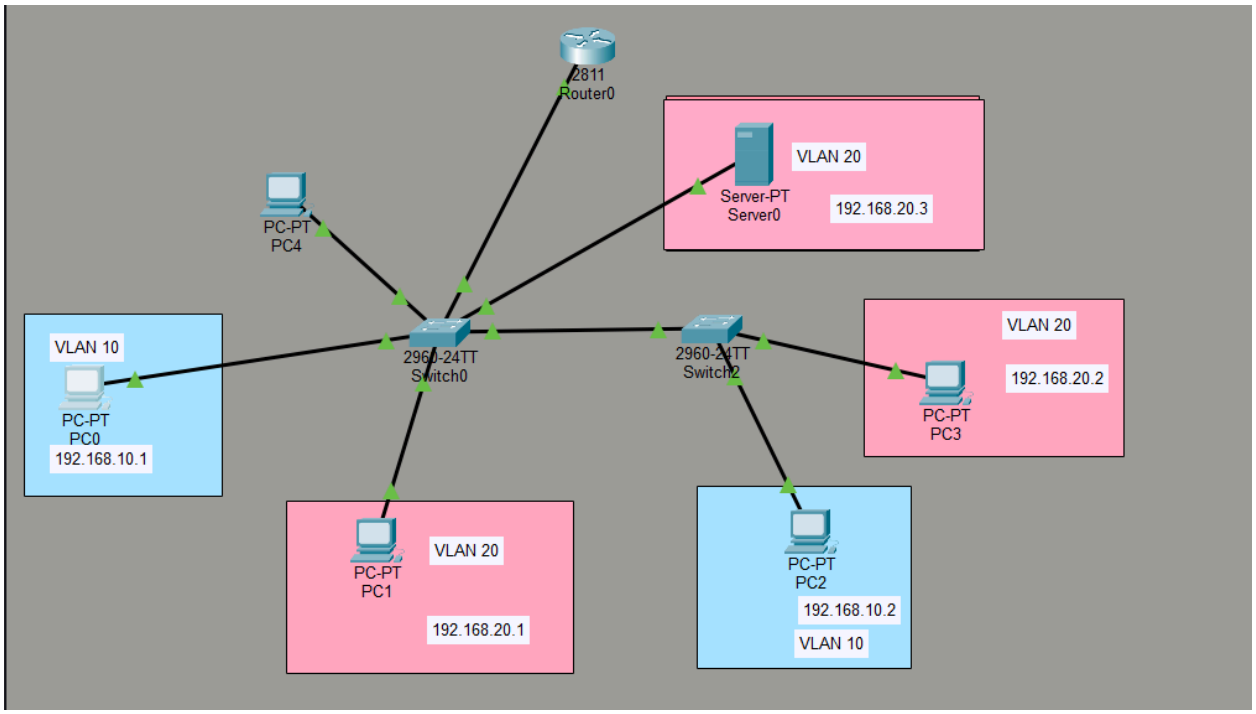
Creating a access-list for https and ftp

```
Router(config-ext-nacl)#ip access-list extended BLOCK_APLS
Router(config-ext-nacl)#deny tcp any any eq 80
Router(config-ext-nacl)#deny tcp any any eq 21
Router(config-ext-nacl)#permit ip any any
Router(config-ext-nacl)#exit
```

Adding this into the interface of VLAN 10 of router

```
Router(config)#interface Fa0/0.10
Router(config-subif)#ip access-group BLOCK_APLS in
Router(config-subif)#exit
```

Connected a server on VLAN20 from https and ftp

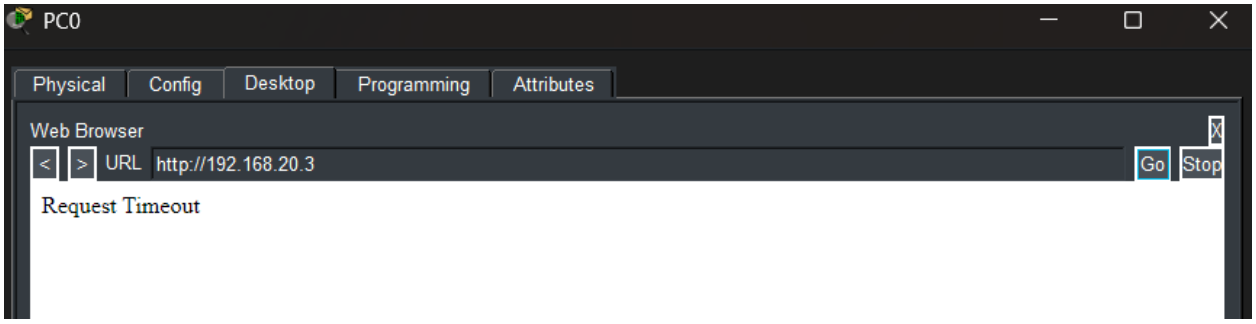



```
SW0(config)#interface FastEthernet0/3
SW0(config-if)#switchport mode access
SW0(config-if)#switchport access vlan20
                                      ^
% Invalid input detected at '^' marker.

SW0(config-if)#switchport access vlan 20
SW0(config-if)#exit
SW0(config)#exit
```

Trying ping from VLAN 10 to server

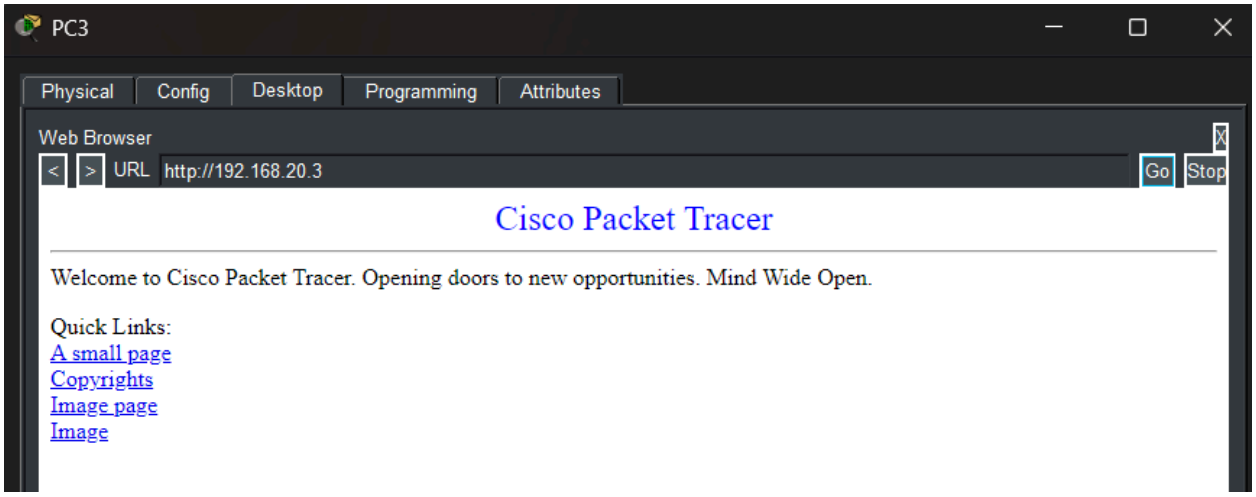
```
C:\>ping 192.168.20.3

Pinging 192.168.20.3 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127
Reply from 192.168.20.3: bytes=32 time<1ms TTL=127

Ping statistics for 192.168.20.3:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

Trying http and ftp from VLAN 10




```
C:\>ftp 192.168.20.3
Trying to connect...192.168.20.3

%Error opening ftp://192.168.20.3/ (Timed out)
.



(Disconnecting from ftp server)
```

Trying http and ftp from VLAN 20




```
C:\>ftp 192.168.20.3
Trying to connect...192.168.20.3
Connected to 192.168.20.3
220- Welcome to PT Ftp server
Username:vimal
331- Username ok, need password
Password:***
230- Logged in
(passive mode On)
ftp>
```

14. Try Static NAT, Dynamic NAT and PAT to translate IPs

Configuring 192.168.10.1 in inside to 203.0.113.1 on outside

```
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 203.0.113.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up
exit
Router(config)#interface FastEthernet0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip nat outside
Router(config-if)#exit
Router(config)#ip nat inside source static 192.168.10.1 203.0.113.3
```

Adding a new router say ISP router and giving it a different IP

```
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 203.0.113.2 255.255.255.0
Router(config-if)#no shutdown
```

Verifying static NAT using pinging and checking translations

```
C:\>ping 203.0.113.2

Pinging 203.0.113.2 with 32 bytes of data:

Reply from 203.0.113.2: bytes=32 time=9ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254

Ping statistics for 203.0.113.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 9ms, Average = 2ms
```

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  203.0.113.3       192.168.10.1       ---                ---
```

Verifying dynamic NAT using pinging and checking translations

Here we create a pool of IP addresses and create ACL to define which device can access NAT

```
Router(config)#ip nat pool PUBLIC_POOL 203.0.113.3 203.0.113.6 netmask 255.255.255.0
Router(config)#ip nat inside source list 11 pool PUBLIC_POOL
```

```
C:\>ping 203.0.113.2

Pinging 203.0.113.2 with 32 bytes of data:

Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254

Ping statistics for 203.0.113.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
---  203.0.113.3       192.168.10.1       ---                ---
```

Configuring and Verifying PAT uusing pinging and checking translations

```
Router(config)#interface FastEthernet0/0.10
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#interface FastEthernet0/0.20
Router(config-subif)#ip nat inside
Router(config-subif)#exit
Router(config)#ip nat inside source list 11 interface FastEthernet0/1 overload
Router(config)#
Router(config)#exit
```

```
Router#show ip nat statistics
Total translations: 7 (0 static, 7 dynamic, 7 extended)
Outside Interfaces: FastEthernet0/1
Inside Interfaces: FastEthernet0/0.10 , FastEthernet0/0.20
Hits: 7  Misses: 843
Expired translations: 0
Dynamic mappings:
Router#show ip nat translations
Pro  Inside global     Inside local       Outside local      Outside global
icmp 203.0.113.1:533   192.168.10.1:533   203.0.113.2:533    203.0.113.2:533
icmp 203.0.113.1:534   192.168.10.1:534   203.0.113.2:534    203.0.113.2:534
icmp 203.0.113.1:535   192.168.10.1:535   203.0.113.2:535    203.0.113.2:535
icmp 203.0.113.1:536   192.168.10.1:536   203.0.113.2:536    203.0.113.2:536
icmp 203.0.113.1:537   192.168.10.1:537   203.0.113.2:537    203.0.113.2:537
icmp 203.0.113.1:538   192.168.10.1:538   203.0.113.2:538    203.0.113.2:538
icmp 203.0.113.1:539   192.168.10.1:539   203.0.113.2:539    203.0.113.2:539
icmp 203.0.113.1:540   192.168.10.1:540   203.0.113.2:540    203.0.113.2:540
icmp 203.0.113.1:541   192.168.10.1:541   203.0.113.2:541    203.0.113.2:541
icmp 203.0.113.1:542   192.168.10.1:542   203.0.113.2:542    203.0.113.2:542
icmp 203.0.113.1:543   192.168.10.1:543   203.0.113.2:543    203.0.113.2:543
icmp 203.0.113.1:544   192.168.10.1:544   203.0.113.2:544    203.0.113.2:544
```

```
C:\>ping 203.0.113.2 -n 100

Pinging 203.0.113.2 with 32 bytes of data:

Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time=1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
Reply from 203.0.113.2: bytes=32 time<1ms TTL=254
```

15. Download iperf in laptop/phone and make sure they are in same network. Try different iperf commands with tcp, udp, birectional, reverse, multicast, parallel options and analyze the bandwidth and rate of transmission, delay, jitter etc.

I don’t have an android to install iperf3, so I am using loopback on my laptop itself for one to act as server and another one as client

Trying tcp test

Output of server

```
C:\Users\vimal\iperf3>iperf3 -s
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 127.0.0.1, port 54960
[  5] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 54961
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-1.00   sec   590 MBytes  4.95 Gbits/sec
[  5]   1.00-2.00   sec   551 MBytes  4.62 Gbits/sec
[  5]   2.00-3.00   sec   563 MBytes  4.73 Gbits/sec
[  5]   3.00-4.00   sec   486 MBytes  4.07 Gbits/sec
[  5]   4.00-5.00   sec   591 MBytes  4.96 Gbits/sec
[  5]   5.00-6.00   sec   469 MBytes  3.93 Gbits/sec
[  5]   6.00-7.00   sec   519 MBytes  4.35 Gbits/sec
[  5]   7.00-8.00   sec   662 MBytes  5.56 Gbits/sec
[  5]   8.00-9.00   sec   689 MBytes  5.78 Gbits/sec
[  5]   9.00-10.00  sec   554 MBytes  4.64 Gbits/sec
[  5]  10.00-10.00  sec  0.00 Bytes  0.00 bits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec                  sender
[  5]   0.00-10.00  sec  5.54 GBytes  4.76 Gbits/sec                  receiver
-----------------------------------------------------------
```

Output of client

```
C:\Users\vimal\iperf3>iperf3 -c 127.0.0.1
Connecting to host 127.0.0.1, port 5201
[  4] local 127.0.0.1 port 54961 connected to 127.0.0.1 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec   590 MBytes  4.95 Gbits/sec
[  4]   1.00-2.00   sec   551 MBytes  4.62 Gbits/sec
[  4]   2.00-3.00   sec   563 MBytes  4.72 Gbits/sec
[  4]   3.00-4.00   sec   486 MBytes  4.07 Gbits/sec
[  4]   4.00-5.00   sec   591 MBytes  4.96 Gbits/sec
[  4]   5.00-6.00   sec   469 MBytes  3.93 Gbits/sec
[  4]   6.00-7.00   sec   519 MBytes  4.35 Gbits/sec
[  4]   7.00-8.00   sec   662 MBytes  5.56 Gbits/sec
[  4]   8.00-9.00   sec   689 MBytes  5.78 Gbits/sec
[  4]   9.00-10.00  sec   554 MBytes  4.65 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  5.54 GBytes  4.76 Gbits/sec                  sender

[  4]   0.00-10.00  sec  5.54 GBytes  4.76 Gbits/sec                  receiver

iperf Done.
```

Analysis

Total data transferred: **5.54 GB** in 10 seconds

Average bandwidth: **4.76 Gbits/sec**

Sender and receiver bandwidth are identical, no packet loss over loopback
Bandwidth fluctuates per second (3.93 to 5.78 Gbits/sec), TCP adjusts its sending rate dynamically using congestion control

Trying UDP Test

Output of server

```
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 127.0.0.1, port 53382
[  5] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 55575
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[  5]   0.00-1.00   sec  9.79 MBytes  82.0 Mbits/sec  0.024 ms  131/1384 (9.5%)
[  5]   1.00-2.00   sec  10.5 MBytes  88.1 Mbits/sec  0.009 ms  179/1523 (12%)
[  5]   2.00-3.00   sec  10.9 MBytes  91.0 Mbits/sec  0.003 ms  138/1527 (9%)
[  5]   3.00-4.00   sec  10.6 MBytes  89.2 Mbits/sec  0.011 ms  168/1528 (11%)
[  5]   4.00-5.00   sec  10.8 MBytes  90.6 Mbits/sec  0.015 ms  143/1524 (9.4%)
[  5]   5.00-6.00   sec  10.3 MBytes  86.3 Mbits/sec  0.016 ms  209/1527 (14%)
[  5]   6.00-7.00   sec  10.5 MBytes  88.4 Mbits/sec  0.003 ms  176/1524 (12%)
[  5]   7.00-8.00   sec  10.6 MBytes  89.3 Mbits/sec  0.024 ms  163/1526 (11%)
[  5]   8.00-9.00   sec  10.6 MBytes  88.7 Mbits/sec  0.015 ms  174/1528 (11%)
[  5]   9.00-10.00  sec  10.2 MBytes  85.4 Mbits/sec  0.005 ms  220/1523 (14%)
[  5]  10.00-10.00  sec  0.00 Bytes  0.00 bits/sec  0.005 ms  0/0 (nan%)
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[  5]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec  0.005 ms  1701/15114 (11%)
```

Output of client

```
C:\Users\vimal\iperf3>iperf3 -c 127.0.0.1 -u -b 100M
Connecting to host 127.0.0.1, port 5201
[  4] local 127.0.0.1 port 55575 connected to 127.0.0.1 port 5201
[ ID] Interval           Transfer     Bandwidth       Total Datagrams

[  4]   0.00-1.00   sec  10.8 MBytes  90.6 Mbits/sec  1382
[  4]   1.00-2.00   sec  11.9 MBytes  99.8 Mbits/sec  1525
[  4]   2.00-3.00   sec  11.9 MBytes   100 Mbits/sec  1527
[  4]   3.00-4.00   sec  11.9 MBytes   100 Mbits/sec  1526
[  4]   4.00-5.00   sec  11.9 MBytes  99.9 Mbits/sec  1526
[  4]   5.00-6.00   sec  11.9 MBytes   100 Mbits/sec  1527
[  4]   6.00-7.00   sec  11.9 MBytes  99.9 Mbits/sec  1524
[  4]   7.00-8.00   sec  11.9 MBytes   100 Mbits/sec  1528
[  4]   8.00-9.00   sec  11.9 MBytes   100 Mbits/sec  1526
[  4]   9.00-10.00  sec  11.9 MBytes  99.9 Mbits/sec  1523
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth       Jitter    Lost/Total Datagrams
[  4]   0.00-10.00  sec   118 MBytes  99.0 Mbits/sec  0.005 ms  1701/15114 (11%)
[  4] Sent 15114 datagrams

iperf Done.
```

Analysis

Target bandwidth: 100 Mbps, achieved **99.0 Mbits/sec** average
Total datagrams sent: **15114**, lost: **1701 (11% packet loss)**
Jitter: **0.005 ms**, low since it is loopback

Trying reverse test

Server output

```
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 127.0.0.1, port 52749
[  5] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 52750
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-1.00   sec  1.34 GBytes  11.5 Gbits/sec
[  5]   1.00-2.00   sec  1.00 GBytes  8.63 Gbits/sec
[  5]   2.00-3.00   sec  1.37 GBytes  11.8 Gbits/sec
[  5]   3.00-4.00   sec  1.44 GBytes  12.4 Gbits/sec
[  5]   4.00-5.00   sec  1.06 GBytes  9.09 Gbits/sec
[  5]   5.00-6.00   sec  1.05 GBytes  9.00 Gbits/sec
[  5]   6.00-7.00   sec  1.16 GBytes  9.99 Gbits/sec
[  5]   7.00-8.00   sec  1.20 GBytes  10.3 Gbits/sec
[  5]   8.00-9.00   sec  1.03 GBytes  8.88 Gbits/sec
[  5]   9.00-10.00  sec  1013 MBytes  8.50 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-10.00  sec  11.6 GBytes  10.0 Gbits/sec                  sender
[  5]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec                  receiver
-----------------------------------------------------------
```

Client output

```
C:\Users\vimal\iperf3>iperf3 -c 127.0.0.1 -R
Connecting to host 127.0.0.1, port 5201
Reverse mode, remote host 127.0.0.1 is sending
[  4] local 127.0.0.1 port 52750 connected to 127.0.0.1 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec  1.34 GBytes  11.5 Gbits/sec
[  4]   1.00-2.00   sec  1.00 GBytes  8.63 Gbits/sec
[  4]   2.00-3.00   sec  1.37 GBytes  11.8 Gbits/sec
[  4]   3.00-4.00   sec  1.44 GBytes  12.4 Gbits/sec
[  4]   4.00-5.00   sec  1.06 GBytes  9.09 Gbits/sec
[  4]   5.00-6.00   sec  1.05 GBytes  9.00 Gbits/sec
[  4]   6.00-7.00   sec  1.16 GBytes  9.99 Gbits/sec
[  4]   7.00-8.00   sec  1.20 GBytes  10.3 Gbits/sec
[  4]   8.00-9.00   sec  1.03 GBytes  8.88 Gbits/sec
[  4]   9.00-10.00  sec  1013 MBytes  8.50 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  11.6 GBytes  10.0 Gbits/sec                  sender
[  4]   0.00-10.00  sec  11.6 GBytes  10.0 Gbits/sec                  receiver

iperf Done.
```

Analysis

Total transferred: 11.6 GB in 10 seconds

Average bandwidth: 10.0 Gbits/sec, more than double the normal TCP test (4.76 Gbits/sec) because in reverse mode the server sends without waiting for requests, so throughput is higher

Trying parallel streams

Server output

```
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 127.0.0.1, port 50909
[  5] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 50910
[  7] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 50911
[  9] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 50912
[ 11] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 50913
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-1.00   sec   734 MBytes  6.16 Gbits/sec
[  7]   0.00-1.00   sec   727 MBytes  6.10 Gbits/sec
[  9]   0.00-1.00   sec   686 MBytes  5.75 Gbits/sec
[ 11]   0.00-1.00   sec   625 MBytes  5.24 Gbits/sec
[SUM]   0.00-1.00   sec  2.71 GBytes  23.2 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[  5]   1.00-2.00   sec   734 MBytes  6.16 Gbits/sec
[  7]   1.00-2.00   sec   713 MBytes  5.98 Gbits/sec
[  9]   1.00-2.00   sec   691 MBytes  5.79 Gbits/sec
[ 11]   1.00-2.00   sec   603 MBytes  5.06 Gbits/sec
[SUM]   1.00-2.00   sec  2.68 GBytes  23.0 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[  5]   2.00-3.00   sec   749 MBytes  6.29 Gbits/sec
[  7]   2.00-3.00   sec   710 MBytes  5.95 Gbits/sec
[  9]   2.00-3.00   sec   680 MBytes  5.70 Gbits/sec
[ 11]   2.00-3.00   sec   608 MBytes  5.10 Gbits/sec
[SUM]   2.00-3.00   sec  2.68 GBytes  23.0 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[  5]   3.00-4.00   sec   799 MBytes  6.70 Gbits/sec
[  7]   3.00-4.00   sec   727 MBytes  6.10 Gbits/sec
[  9]   3.00-4.00   sec   680 MBytes  5.71 Gbits/sec
[ 11]   3.00-4.00   sec   614 MBytes  5.15 Gbits/sec
[SUM]   3.00-4.00   sec  2.75 GBytes  23.7 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
```

```
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec                  sender
[  5]   0.00-10.00  sec  7.35 GBytes  6.31 Gbits/sec                  receiver
[  7]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec                  sender
[  7]   0.00-10.00  sec  6.97 GBytes  5.99 Gbits/sec                  receiver
[  9]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec                  sender
[  9]   0.00-10.00  sec  6.66 GBytes  5.72 Gbits/sec                  receiver
[ 11]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec                  sender
[ 11]   0.00-10.00  sec  5.88 GBytes  5.05 Gbits/sec                  receiver
[SUM]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec                  sender
[SUM]   0.00-10.00  sec  26.9 GBytes  23.1 Gbits/sec                  receiver
```

Client output

```
C:\Users\vimal\iperf3>iperf3 -c 127.0.0.1 -P 4
Connecting to host 127.0.0.1, port 5201
[  4] local 127.0.0.1 port 50910 connected to 127.0.0.1 port 5201
[  6] local 127.0.0.1 port 50911 connected to 127.0.0.1 port 5201
[  8] local 127.0.0.1 port 50912 connected to 127.0.0.1 port 5201
[ 10] local 127.0.0.1 port 50913 connected to 127.0.0.1 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec   734 MBytes  6.16 Gbits/sec
[  6]   0.00-1.00   sec   727 MBytes  6.10 Gbits/sec
[  8]   0.00-1.00   sec   686 MBytes  5.76 Gbits/sec
[ 10]   0.00-1.00   sec   625 MBytes  5.24 Gbits/sec
[SUM]   0.00-1.00   sec  2.71 GBytes  23.3 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - - -
[  4]   1.00-2.00   sec   734 MBytes  6.16 Gbits/sec
[  6]   1.00-2.00   sec   713 MBytes  5.98 Gbits/sec
[  8]   1.00-2.00   sec   691 MBytes  5.79 Gbits/sec
[ 10]   1.00-2.00   sec   603 MBytes  5.05 Gbits/sec
[SUM]   1.00-2.00   sec  2.68 GBytes  23.0 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - - -
[  4]   2.00-3.00   sec   749 MBytes  6.29 Gbits/sec
[  6]   2.00-3.00   sec   710 MBytes  5.95 Gbits/sec
[  8]   2.00-3.00   sec   680 MBytes  5.70 Gbits/sec
[ 10]   2.00-3.00   sec   608 MBytes  5.10 Gbits/sec
[SUM]   2.00-3.00   sec  2.68 GBytes  23.0 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - - -
[  4]   3.00-4.00   sec   799 MBytes  6.70 Gbits/sec
[  6]   3.00-4.00   sec   728 MBytes  6.10 Gbits/sec
[  8]   3.00-4.00   sec   680 MBytes  5.71 Gbits/sec
[ 10]   3.00-4.00   sec   614 MBytes  5.15 Gbits/sec
[SUM]   3.00-4.00   sec  2.76 GBytes  23.7 Gbits/sec
```

```
- - - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  7.35 GBytes  6.31 Gbits/sec                  sender
[  4]   0.00-10.00  sec  7.35 GBytes  6.31 Gbits/sec                  receiver
[  6]   0.00-10.00  sec  6.97 GBytes  5.99 Gbits/sec                  sender
[  6]   0.00-10.00  sec  6.97 GBytes  5.99 Gbits/sec                  receiver
[  8]   0.00-10.00  sec  6.66 GBytes  5.72 Gbits/sec                  sender
[  8]   0.00-10.00  sec  6.66 GBytes  5.72 Gbits/sec                  receiver
[ 10]   0.00-10.00  sec  5.88 GBytes  5.05 Gbits/sec                  sender
[ 10]   0.00-10.00  sec  5.88 GBytes  5.05 Gbits/sec                  receiver
[SUM]   0.00-10.00  sec  26.9 GBytes  23.1 Gbits/sec                  sender
[SUM]   0.00-10.00  sec  26.9 GBytes  23.1 Gbits/sec                  receiver
```

Analysis

- 4 simultaneous TCP streams opened (ports 50910, 50911, 50912, 50913)
- Each stream bandwidth individually: 5.05 to 6.31 Gbits/sec
- Combined total (SUM): 23.1 Gbits/sec, nearly 5x the single stream speed
- Total data transferred: 26.9 GB in 10 seconds
- All 4 streams show zero packet loss — TCP delivered everything reliably

Trying bi-directional streams

Server output

```
-----------------------------------------------------------
Server listening on 5201
-----------------------------------------------------------
Accepted connection from 127.0.0.1, port 58014
[  5] local 127.0.0.1 port 5201 connected to 127.0.0.1 port 58015
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-1.00   sec   579 MBytes  4.85 Gbits/sec
[  5]   1.00-2.00   sec   653 MBytes  5.48 Gbits/sec
[  5]   2.00-3.00   sec   717 MBytes  6.02 Gbits/sec
[  5]   3.00-4.00   sec   649 MBytes  5.44 Gbits/sec
[  5]   4.00-5.00   sec   691 MBytes  5.80 Gbits/sec
[  5]   5.00-6.00   sec   630 MBytes  5.28 Gbits/sec
[  5]   6.00-7.00   sec   725 MBytes  6.08 Gbits/sec
[  5]   7.00-8.00   sec   744 MBytes  6.24 Gbits/sec
[  5]   8.00-9.00   sec   584 MBytes  4.90 Gbits/sec
[  5]   9.00-10.00  sec   638 MBytes  5.36 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  5]   0.00-10.00  sec  0.00 Bytes  0.00 bits/sec
   sender
[  5]   0.00-10.00  sec  6.46 GBytes  5.54 Gbits/sec
     receiver
-----------------------------------------------------------
Server listening on 5201
```

Client output

```
C:\Users\vimal\iperf3>iperf3 -c 127.0.0.1 -d
send_parameters:
{
        "tcp":  true,
        "omit": 0,
        "time": 10,
        "parallel":     1,
        "len":  131072,
        "client_version":       "3.1.1"
}
Connecting to host 127.0.0.1, port 5201
SO_SNDBUF is 64512
[  4] local 127.0.0.1 port 58015 connected to 127.0.0.1 port 5201
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-1.00   sec   579 MBytes  4.85 Gbits/sec
[  4]   1.00-2.00   sec   653 MBytes  5.48 Gbits/sec
[  4]   2.00-3.00   sec   717 MBytes  6.02 Gbits/sec
[  4]   3.00-4.00   sec   649 MBytes  5.44 Gbits/sec
[  4]   4.00-5.00   sec   691 MBytes  5.80 Gbits/sec
[  4]   5.00-6.00   sec   630 MBytes  5.28 Gbits/sec
[  4]   6.00-7.00   sec   725 MBytes  6.08 Gbits/sec
[  4]   7.00-8.00   sec   744 MBytes  6.24 Gbits/sec
[  4]   8.00-9.00   sec   584 MBytes  4.90 Gbits/sec
send_results
{
        "cpu_util_total":       31.0718,
        "cpu_util_user":        2.80836,
        "cpu_util_system":      28.2635,
        "sender_has_retransmits":       0,
        "streams":      [{
                        "id":   1,
                        "bytes":        6931262464,
                        "retransmits":  -1,
                        "jitter":       0,
```

```
                        "errors":       0,
                        "packets":      0
                }]
}
get_results
{
        "cpu_util_total":       0.997427,
        "cpu_util_user":        0.113447,
        "cpu_util_system":      0.88398,
        "sender_has_retransmits":       -1,
        "streams":      [{
                        "id":   1,
                        "bytes":        6931004416,
                        "retransmits":  -1,
                        "jitter":       0,
                        "errors":       0,
                        "packets":      0
                }]
}
[  4]   9.00-10.00  sec   638 MBytes  5.36 Gbits/sec
- - - - - - - - - - - - - - - - - - - - - - - - -
[ ID] Interval           Transfer     Bandwidth
[  4]   0.00-10.00  sec  6.46 GBytes  5.54 Gbits/sec                  sender
[  4]   0.00-10.00  sec  6.46 GBytes  5.54 Gbits/sec                  receiver

iperf Done.
```

Analysis

- Total transferred: 6.46 GB in 10 seconds
- Average bandwidth: 5.54 Gbits/sec
- Retransmits: 0, perfect TCP delivery

- Zero packet loss and zero errors
- CPU usage on
    - Sender: 31%
    - Receiver: 1%